

GCE ADVANCED LEVEL STUDENT PERFORMANCE DATA WAREHOUSE PROJECT REPORT

Student Performance (GCE AL Exam 2020 - Sri Lanka)

Data Source: Kaggle

ETL Platform: Google Colab, Python and Pandas

Data Warehouse: PostgreSQL 18

Database Management Tool: pgAdmin

Project Summary

This project develops a dimensional Data Warehouse for analysing Sri Lankan GCE Advanced Level examination performance data from 2020. The raw Kaggle dataset was extracted, cleaned and transformed using Python and Pandas in Google Colab and subsequently loaded into a PostgreSQL Data Warehouse using a Star Schema.

Prepared for Academic Data Warehousing Study

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1. Introduction

A Data Warehouse is a centralized repository designed primarily for analysis, reporting and decision making. Unlike an operational database that focuses on day-to-day transactions, a Data Warehouse organizes historical data so that users can analyse information from multiple perspectives.

This project develops a Data Warehouse using the Sri Lankan GCE Advanced Level examination dataset for the year 2020. The objective is to transform the original flat CSV dataset into a dimensional model that supports analytical queries related to student performance.

The implemented solution uses Google Colab with Python and Pandas for the Extract and Transform stages and PostgreSQL 18 with pgAdmin for the Load stage.

2. Dataset Description

The source dataset is the Kaggle dataset titled **Student Performance (GCE AL Exam_2020 - Sri Lanka)**. The dataset contains examination-related information for students who participated in the 2020 Sri Lankan GCE Advanced Level examination.

Property	Value
Source	Kaggle
Examination Year	2020
Original Records	337,553
Original Columns	19
Subject Records after transformation	1,012,659

Main Source Attributes

Column	Description
index	Original student record identifier
stream	A/L study stream
Zscore	Student Z-score
district_rank	District examination rank
island_rank	Island-wide examination rank
al_year	Examination year
sub1 / sub1_r	First subject and grade
sub2 / sub2_r	Second subject and grade
sub3 / sub3_r	Third subject and grade
cgt_r	Common General Test result

Column	Description
ge_r	General English result
syllabus	New or old syllabus
birth_day/month/year	Student birth date components
gender	Student gender

3. Data Warehouse Analytical Requirements

The Data Warehouse was designed around the analytical questions that can be answered using the student performance data. The main analytical requirements are:

1. Analyse examination performance by subject.
2. Compare performance between different A/L streams.
3. Analyse student performance by gender.
4. Compare performance between new and old syllabi.
5. Analyse grade distributions for individual subjects.
6. Analyse Z-score performance.
7. Analyse district and island rankings.
8. Support multidimensional reporting and decision making.

4. ETL Process

4.1 Extract

The original student performance CSV file was obtained from Kaggle. The dataset was loaded into Google Colab using Python and Pandas. The initial dataset contained 337,553 records and 19 columns.

4.2 Transform

The transformation stage prepared the raw data for dimensional modelling. Z-score values and ranking fields were converted into numeric values. The separate birth day, month and year fields were combined into a single birth_date field.

The three subject columns were also reshaped. The original structure stored three subjects horizontally in each student record. These were converted into separate subject-performance rows. As a result, each student can have three records in the fact table.

4.3 Load

After transformation, the dimension and fact tables were exported as CSV files from Google Colab. These CSV files were then imported into PostgreSQL using pgAdmin.

ETL Flow: Kaggle CSV → Python/Pandas → Cleaned Data → Dimensional Model → CSV Files → PostgreSQL Data Warehouse

5. Data Cleaning and Transformation

Transformation	Purpose
Convert Z-score to numeric	Enable numerical analysis
Convert district rank to integer	Match PostgreSQL data type
Convert island rank to integer	Match PostgreSQL data type
Combine birth date fields	Create a proper DATE value
Clean text fields	Remove unwanted spaces and normalize values
Reshape subject columns	Create one student-subject fact record
Handle missing values	Preserve source information without inventing values

During the PostgreSQL loading stage, two data-type issues were identified and corrected. Rank values such as 4336.0 were converted to integers because the database columns were defined as INTEGER. The grade value 'Absent' was also identified; the grade column was therefore expanded from VARCHAR(5) to VARCHAR(10).

6. Data Warehouse Architecture

The implemented Data Warehouse follows a dimensional architecture. The central fact table contains student-subject performance, while five dimension tables provide descriptive context for analysis.

Component	Technology / Purpose
Source	Kaggle CSV dataset
Extract & Transform	Google Colab + Python + Pandas
Data Warehouse	PostgreSQL 18
Database Interface	pgAdmin
Schema Type	Star Schema
Fact Table	fact_student_subject
Dimension Tables	5 dimensions

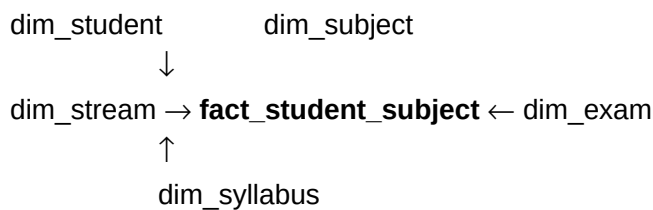
7. Star Schema Design

A Star Schema is a Data Warehouse modelling technique in which a central fact table is directly connected to several dimension tables. The model resembles a star because the fact table is located at the centre and the dimensions surround it.

Implemented Star Schema:

Dimension Table	Relationship	Central Fact Table
dim_student	student_key	fact_student_subject
dim_subject	subject_key	fact_student_subject
dim_stream	stream_key	fact_student_subject
dim_exam	exam_key	fact_student_subject
dim_syllabus	syllabus_key	fact_student_subject

Simple representation:



8. Fact Table

fact_student_subject

The fact table is the central table in the Data Warehouse. Its grain is **one row per student per subject performance**. Since the source dataset contains three subjects for each student, the resulting fact table contains 1,012,659 records.

Column	Purpose
fact_key	Unique fact record identifier
student_key	Links to dim_student
subject_key	Links to dim_subject
stream_key	Links to dim_stream
exam_key	Links to dim_exam
syllabus_key	Links to dim_syllabus
z_score	Student Z-score
district_rank	District rank
island_rank	Island-wide rank
grade	Subject grade/result

Fact table size: 1,012,659 records

9. Dimension Tables

Dimension	Rows	Purpose
dim_student	337,553	Student descriptive information
dim_subject	62	Subject information
dim_stream	8	A/L stream information
dim_exam	1	Examination year
dim_syllabus	2	New / old syllabus

dim_student

Stores the student dimension using a surrogate student_key together with the original student identifier, gender and birth date.

dim_subject

Stores unique subject names and assigns a subject_key to each subject.

dim_stream

Stores the different A/L streams found in the dataset, including Arts, Commerce, Physical Science and other source categories.

dim_exam

Stores the examination year. In this dataset there is one examination year, 2020.

dim_syllabus

Stores the syllabus type, which consists of new and old syllabus categories.

10. PostgreSQL Implementation

PostgreSQL 18 was selected as the Data Warehouse database management system. pgAdmin was used as the graphical interface for creating tables, executing SQL queries and importing the transformed CSV files.

Main Tables Created

The database contains six tables: five dimension tables and one central fact table.

Primary keys were created for all dimension tables and the fact table. Foreign key constraints were created in the fact table to maintain referential integrity between the fact and dimension tables.

11. Data Loading and Validation

The transformed CSV files were loaded into PostgreSQL in the following order: dimension tables first and the fact table afterward. Loading the dimensions first ensures that the foreign key references required by the fact table already exist.

Table	Expected / Loaded Rows	Status
dim_student	337,553	Successful
dim_subject	62	Successful
dim_stream	8	Successful
dim_exam	1	Successful
dim_syllabus	2	Successful
fact_student_subject	1,012,659	Successful

The final row-count validation confirmed that all six tables were loaded successfully and that the fact table contains 1,012,659 records.

12. Analytical SQL Queries

12.1 Average Z-score by Stream

This query demonstrates how the fact table can be combined with the stream dimension to calculate average student performance by stream.

```
SELECT s.stream_name,  
AVG(f.z_score) AS average_z_score  
FROM fact_student_subject f  
JOIN dim_stream s ON f.stream_key = s.stream_key  
GROUP BY s.stream_name  
ORDER BY average_z_score DESC;
```

12.2 Average Z-score by Subject

```
SELECT s.subject_name,  
AVG(f.z_score) AS average_z_score  
FROM fact_student_subject f  
JOIN dim_subject s ON f.subject_key = s.subject_key  
GROUP BY s.subject_name  
ORDER BY average_z_score DESC;
```

12.3 Grade Distribution

```
SELECT s.subject_name,  
f.grade,  
COUNT(*) AS student_count  
FROM fact_student_subject f  
JOIN dim_subject s ON f.subject_key = s.subject_key  
GROUP BY s.subject_name, f.grade  
ORDER BY s.subject_name, student_count DESC;
```

12.4 Performance by Gender

```
SELECT st.gender,  
AVG(f.z_score) AS average_z_score  
FROM fact_student_subject f  
JOIN dim_student st ON f.student_key = st.student_key
```



```
GROUP BY st.gender;
```

12.5 New vs Old Syllabus

```
SELECT sy.syllabus_name,  
AVG(f.z_score) AS average_z_score  
FROM fact_student_subject f  
JOIN dim_syllabus sy ON f.syllabus_key = sy.syllabus_key  
GROUP BY sy.syllabus_name;
```

13. Example Analytical Results

The Data Warehouse enables a range of analytical reports. The following are examples of the types of findings that can be produced from the warehouse.

Analysis	Purpose
Average Z-score by stream	Compare overall performance between streams
Average Z-score by subject	Identify relative subject performance
Grade distribution	Understand grades achieved in each subject
Gender analysis	Compare performance between genders
Syllabus comparison	Compare new and old syllabus performance
Ranking analysis	Study district and island rankings

These analytical queries demonstrate the main purpose of the Data Warehouse: converting raw examination data into structured information that can support analysis and reporting.

14. Why Star Schema Was Selected

The Star Schema was selected because it matches the analytical requirements of this project. The main requirement is to analyse student performance from multiple perspectives, including subject, stream, gender, examination year and syllabus.

The fact table contains the performance information, while the dimension tables provide the different perspectives required for analysis.

A Snowflake Schema could also be used, but it would further normalize dimension data and introduce additional tables and joins. The dimensions in this project are relatively simple, so that additional complexity is not necessary.

Therefore, Star Schema was chosen because it provides:

1. Simple and easy-to-understand structure.
2. Fewer joins for analytical queries.
3. Good suitability for reporting and OLAP-style analysis.
4. Easy navigation between facts and dimensions.

5. A clear separation between measurable facts and descriptive dimensions.
6. A structure that directly matches the project's analytical requirements.

15. Data Warehouse Advantages

1. Centralizes examination data into a structured analytical repository.
2. Makes analytical queries easier than working directly with the original CSV.
3. Separates facts from descriptive dimensions.
4. Supports multidimensional analysis.
5. Allows aggregation of Z-scores and ranking information.
6. Provides a foundation for dashboards and business intelligence tools.
7. Improves data organization for reporting and decision making.

16. Limitations

There are several limitations in the current implementation.

1. The dataset represents a single examination year, 2020.
2. The source dataset does not provide detailed historical changes.
3. The source does not provide a full district dimension with district names.
4. Missing values exist in some source attributes such as gender and performance fields.
5. The current warehouse focuses on examination performance rather than wider student information.

17. Conclusion

This project successfully transformed the Sri Lankan GCE Advanced Level 2020 student performance dataset into a functional Data Warehouse.

The Extract, Transform and Load process was implemented using Kaggle, Python, Pandas, Google Colab and PostgreSQL. The original 337,553 student records were transformed into a dimensional structure containing five dimensions and one fact table.

The final Data Warehouse contains 337,553 student dimension records, 62 subjects, 8 streams, 1 examination year, 2 syllabus types and 1,012,659 student-subject fact records.

The Star Schema provides a simple and effective structure for analysing student performance by subject, stream, gender, syllabus and other dimensions. The implemented analytical queries demonstrate how the warehouse can support reporting and decision-making.

Overall, the project demonstrates the complete Data Warehouse workflow from raw source data through ETL transformation and database loading to analytical querying.

18. References

Kaggle. Student Performance (GCE AL Exam_2020 - Sri Lanka). Dataset by Sasika Amarasinghe.

PostgreSQL Documentation. PostgreSQL relational database management system documentation.

Pandas Documentation. Python data analysis and manipulation library.

ReportLab Documentation. Python library used to generate this PDF report.